

VERITAS: A Scalable Microservice Architecture for Multi-Modal Manuscript Integrity and Citation Anomaly Detection

Billvin Christo Thomas, Hrishikesh S, KV Chandana Chithran, Liz Maria Benny, Dr Varghese S Chooralil

Dept. of AI and DS

Rajagiri School of Engineering and Technology

Kochi, Kerala, India

btcanad2@gmail.com, chithrankvchandana@gmail.com

,hrishi2604@gmail.com, maria.liz.benny@gmail.com, Varghese@rajagiritech.edu.in

Abstract—

THE increase in the academic output flow has been accompanied by the increased research integrity issues, such as falsified citations, semantic inconsistency, and text by AI. The conventional plagiarism detectors mainly deal with shallow textual resemblances and do not result in significant structural and semantic deviations.

In this paper, we will describe VERITAS, an AI-assisted platform, which is used to analyze research manuscript proactively and derive an interpretable score of the Integrity risk. VERITAS combines citation forensics with semantic consistency checking and AI-authorship recognizer with two fundamental modules: Citation and Stylometry Anomaly Detector (CSAD) and Semantic Integrity and AI-Authorship Verifier (SAIV).

The citation analysis of a single paper is done by CSAD which detects excessive self-citation, citation to retracted publications, atypical citation distributions within sections and stylometric deviations that exhibit patchwork authorship. Local citation structures are analyzed using lightweight graph modeling and can be used to explain risk scoring in an explainable way.

SAIV uses transformer-based language models to check semantic coherence of the sections of the manuals and approximate the chance of AI-written text by the probabilistic classification of the texts that is improved with stylistic properties.

The experimental findings prove that VERITAS is capable of detecting citation abnormalities and semantic anomalies and is still easy to interpret by reviewers. The scalability of the deployment made by the modular architecture makes the platform an efficient decision-support tool to the editors and researchers.

Index Terms—Research Integrity, Citation Analysis, Stylometry, AI-generated Text Detection, Semantic Consistency.

I. INTRODUCTION

The fact that scholarly publications are growing incredibly fast has made the progress of science much faster, however, it has raised new concerns related to the integrity of research. Problems of inconsistencies in the citation styles, irregular format of the manuscript and the growing use of generative artificial intelligence (AI) in the academic writing have complicated quality control among journals and conferences. The recent developments in the retractions of papers show that conventional integrity checking methods are becoming

hard to keep up with the scale and the complexity of modern submissions.

The tools available mainly concentrate on plagiarism by similarity of the text. Although they are good at locating instances of direct copying, they do not tend to identify underlying integrity threats, including, but not limited, abnormal citation patterns, poor semantic correspondence between manuscript segments, or signs of autopilot writing. This forces editors and reviewers to spend a lot of time and time doing the work manually that is time consuming and subjective.

The new methods of natural language processing and data analytics are promising alternatives. Transformer-based models have demonstrated high potential in semantic structure of scientific text, and both bibliometric and stylometric analysis can demonstrate anomalies in citation style and writing style. Nevertheless, these methods are normally used as standalone procedures and are not amalgamated into a single and understandable framework that is compatible with editorial processes.

This paper introduces the concept of the AI-based proactive and interpretable research integrity assessment platform called VERITAS. VERITAS is not truly binary, but rather a primary screening mechanism to provide an explainable score of Integrity Risk by synthesizing signals related to citation behaviour, writing style consistency, semantic coherence, and AI-authorship probability.

VERITAS has two fundamental modules. The Citation and Stylometry Anomaly Detector (CSAD) puts emphasis on paper internal analysis including the analysis of pattern of references, density of self-citation, retracted citations, and stylometric consistency. The Semantic Integrity /AI-Authorship Verifier (SAIV) measures semantic consistency within the sections of a manuscript and approximates the likelihood of AI-written text with representations based on transformers and supervised learning.

The platform is planned to be the modular and scalable web system which enables the editorial and peer-review processes. VERITAS seeks to support human reviewers by pointing

out possible risk areas and deliver insights to interpret, as opposed to expert judgment. Further research will bring the system to include network-level citation analysis and large-scale bibliographic modeling.

II. RELATED WORK

The research integrity evaluation includes citation analysis, bibliometrics, stylometry, and identifying AI-generated text. Although major progress has been achieved in each of the fields, there has been limited integration and mutual assessment systems of these aspects.

A. Citation Anomalies and Citation Cartels

Network-based studies have long been used to determine abnormal citation patterns. Kojaku, Livan, and Masuda improved the methods of identifying aberrant citation networks in journal corpora, showing that the overabundance of inter-journal citations can be a sign of collective action [1]. On the same note, Fister Jr., Fister, and Perc studied the network structure of citation and found that there is the presence of citation cartels and citation manipulation tendencies [12].

Deep-learning has also been applied to refine a detection of anomalies. Liu et al. proposed graph-based learning models in order to detect anomalous citation patterns using contextual information and network properties, thus demonstrating that citation deviations cannot be identified using frequency measures only [2].

There are also empirical studies on manipulation of author-centric citation. Wren and Georgescu found that there were aberrant referencing styles in PubMed, and that reference lists were being manipulated to give preference to specific authors [3]. This piece of writing highlights the significance of identifying excessive self-citation and biased referencing on a manuscript level.

Earlier bibliometrical studies have demonstrated that self-citing can greatly exaggerate the perception of impact [10]. Large standardized citation databases like those characterised by Ioannidis et al., also show how citation metrics are used in academic evaluation systems [11]. Such results prompt a thorough investigation of the distributions of citations in single manuscripts.

B. Retracted Publications and Bibliometric Risk Indicators

Retraction events are salient indicators of integrity risk. Candal-Pedreira et al. studied retracted papers by paper mills and found that systematic wrongdoing was carried out in the features of references and citations [4].

Most recently, Liu, Wang, and Liang used interpretable machine-learning methods to detect bibliometric aspects of retracted biomedical articles, and found that citation-based features could be used to forecast integrity issues [5].

The academic interest in structural and systematic dimensions of misconduct is becoming more explicit, which is reflected in bibliometric studies of academic integrity research, as such (e.g., Maral) [9]. Taken together, these studies indicate the need to integrate citation-related risk factors into the integrity assessment systems.

C. Stylometry and Authorship Analysis

Authorship attribution and writing style assessment have been traditionally done by stylometric analysis. Discrepancies in authorship may be revealed through differences in lexical richness, syntactic patterns, distribution of sentence-length, and disguised content reuse [13]. It has been shown by recent studies that the stylometric features are applied well to differentiate between the human-written texts and the texts produced by the large language models whereby they capture characteristic stylistic features such as grammatical regularities, lexical variety, and punctuation patterns [13].

Even though stylometry as a singular tool is incapable of conclusively showing cases of misconduct, it offers objective, quantifiable points of writing anomalies that can be used in tandem with other metrics of integrity indicators.

The incorporation of stylometric analysis with modern machine-learning techniques can make it more functional, especially in the context of the ability to distinguish between human-written and machine-written text. However, a purely stylistic form of evaluation is vulnerable to the limitations of semantic inconsistency; therefore, the thorough integrity check must incorporate the element of stylistic analysis along with the extensive analysis of the semantics [13].

D. Detection of AI-Generated Scientific Text

Generative AI tools have rapidly been adopted as a result of which the field of AI-text detection has grown through extensive research. According to a study by Desaire et al., AI-generated scientific texts can be used to escape detection in case they are specific to particular fields, and this is an indication of the intricacies of classification [6].

The careful review of AI-text detectors by Weber-Wulff et al. reported a significant degree of consistency of detections with a lot of false positives and false negatives, thus showing that standalone AI-text detectors cannot be relied on to make high-stakes editorial choices [7].

Transformer based language models have enhanced semantic representations of scientific text. Beltagy, Lo and Cohan released SciBERT serving as a source of domain specific embeddings, obtained using scientific corpora, with improved semantic modeling and classification performances [8]. These models can be used to analyze semantic coherence in more depth than similarity measures which are superficial.

E. Research Gap

The existing methodologies mostly work at the scale of global citation networks, [1], [12] or are limited to small scope of AI-text detection, [6], [7] or bibliometric retractions analysis, [4], [5]. Despite this usefulness, these approaches in general have three weaknesses: (1) reliance on large external citation databases; (2) limited by one dimension of detection; and (3) inadequate interpretability to be useful for editorial practice.

The lack of connection between advanced research practices and practical manuscript-level integrity checking software that simultaneously evaluate the use of citation practices, stylistic

uniformity, semantic consistency, and AI-authorship prediction is strong and well within an interpretable context.

VERITAS bridges this gap by incorporating citation anomaly detection, stylometric analysis, transformer based semantic modeling, and AI based likelihood estimation of authorship to a single and explanatory structure of proactive research inquiry on integrity.

III. SYSTEM ARCHITECTURE

VERITAS is developed as a research integrity analysis platform, which is oriented towards modularity, scalability, and explainability. The system has a layered design to ensure user contact, system management and analysis are independent of one another. This division simplifies the maintenance of the system as well as enables new analysis parts to be introduced in future without change of the current workflow.

A. Architectural Overview

VERITAS architecture is divided into four key components, namely the application layer, the backend orchestration layer, the AI/Data Science analysis layer, and the infrastructure layer as indicated in Fig. 1.

Authors and editors are also users who can participate in the system by accessing a single dashboard in which they can upload the manuscript and view the results of the analysis. Once submitted, the backend communicates with various modules of analysis who analyze the manuscript in various viewpoints. The system does not provide a simple decision but instead, creates an Integrity Risk Report, which identifies potential concerns and justifications. This assists editors to decide on the basis of evidence, as opposed to the use of automated judgment only.

B. Application Layer

The front-end interface of VERITAS is the application layer. It deals with user login, manuscript submission and report visualization. Paper works are uploaded as PDF files and sent to the back-end to be processed. The interface primarily aims to enhance the readability of findings and display integrity points, flagged observations and summaries in a systematic way. It is meant to make sure that the users can comprehend the findings without undergoing technical expertise of the models underpinning it.

C. Backend and Orchestration Layer

The server is the central control of the system. Its primary role is to handle document preprocessors, trigger analysis modules and synthesize their results. After receiving a manuscript, text and metadata are retrieved and made available to further analysis. This processed content is then forwarded to the AI modules and their output is normalized after which they are combined into one Integrity Risk Score. This layer also handles storage operations, user permissions and retrieval of past analysis reports.

D. Data Science Analysis Layer

VERITAS has an analytical functionality that is realized in two fundamental modules that run under the AI/Data Science layer.

The project is called Citation and Stylometry Anomaly Detector (CSAD): CSAD compares variations of citation behavior and writing style in a manuscript. It verifies patterns like self-citation rates that are outlier, reference to withdrawn work, citation concentration in small sources, and apparent stylistic differences in sections. These observations are based on the statistical and data-driven approaches with an aim to detect inconsistencies as opposed to imposing strict rules.

Semantic Integrity AI-Authorship Verifier (SAIV): SAIV is concerned with the assessment of semantic consistency and estimating the potential of AI-supported authorship. Embeddings on transformers are utilized to investigate semantic connections of different parts of the manuscript. According to these models, the supervised learning models determine the extent to which the patterns of writing are similar to AI-generated ones.

The modules are independent and present structured outputs of scores and explanatory indicators, which are combined eventually in the backend.

E. Data storage and Infrastructure Layer

Manuscripts can be stored in a dedicated database to enable their future access and audit where generated reports are stored. The system will be deployed in a cloud-based environment and is based on the approach of a microservices-oriented system to allow scalability and reliable performance. A separation between infrastructure and analytical modules is maintained in order to ensure that deployment changes do not have an impact on analysis outcomes.

F. Explainability and Extensibility

Another major design consideration in VERITAS is explainability. The intermediary indicators generated in the process of analysis are retained and reflected in the final report to enable the users to appreciate through which means the system came up with the conclusions. Future extensions, e.g. cross-manuscript comparison or network-based citation analysis, are also possible in the modular structure but the current implementation can only do an accurate analysis of separate manuscripts.

Figure 2 illustrates the end-to-end workflow of the VERITAS platform, beginning with the manuscript submission and going through the modular citation, stylometric, and semantic analysis modules. The architecture focuses on explainability and scalability by isolating analysis modules and consolidating the outputs of the modules in one report of integrity risks.

IV. METHODOLOGY

In this section, the authors relate the methodology used by VERITAS. The platform uses a multi-dimensional strategy of integrity assessment, involving analysis of citation behavior, stylometric consistency analysis, and semantic analysis. The

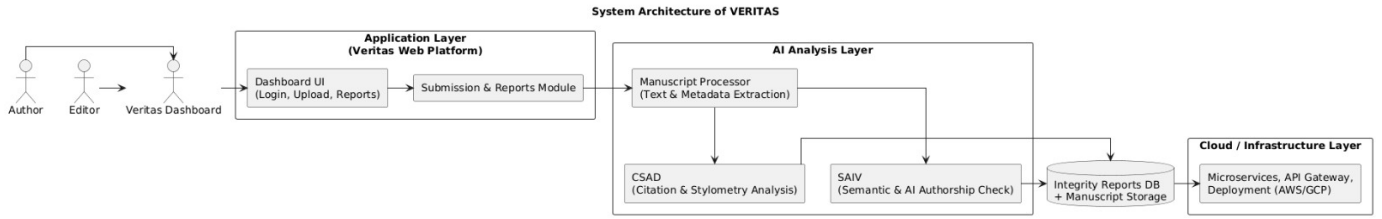


Fig. 1. System architecture of the VERITAS platform showing the layered design and interaction between system components.



Fig. 2. Overall workflow of the VERITAS platform, illustrating manuscript ingestion, orchestration of CSAD and SAIV analysis modules, score aggregation, and generation of an explainable integrity risk report.

methodology is split into two main elements that are related to the CSAD and SAIV modules.

A. Citation and Stylometry Anomaly Detector (CSAD)

The Citation & Stylometry Anomaly Detector (CSAD) is built to detect both paper-internal and reference-local integrity dangers associated with the citation practices and writing style. Instead of being based on the large-scale bibliographic crawling, CSAD solely works based on what is topically and semantically extracted out of a single submitted manuscript, which makes it robust, deployable, and explainable.

Preprocess and extract reference: This is done by extracting valid references from a large body of text according to their relevance to a specific topic.

1) Document Preprocessing and Reference Extraction:

This is performed by retrieving legitimate references in a substantial amount of text based on their applicability to a given subject.

When submitting the manuscript, the PDF document undergoes what is known as metadata processing in order to obtain the structured text and reference metadata. The paper has been divided into rational parts like the abstract, main body, and references. The data points of author names, publication year, venue details and Digital Object Identifiers (DOIs) are detected in reference entries when present. It is a structured representation on which the future citing and stylometric analysis is built.

2) Retracted Reference Detection: CSAD conducts reference integrity checks by detecting the citations to retracted publications. DOIs are extracted and compared to a maintained retraction database. In case a referenced article has been retracted, the source is labeled as a possible integrity threat. References to retracted citations are included in the total risk score of CSAD since they can represent the lack of a proper literature review or purposeful misuse.

3) Self-Citation Density Analysis: CSAD calculates the self-citation ratio, which involves comparison of the list of authors on the manuscript with the authors of referenced sources. The names of the authors are normalized to take care of the differences in formatting. A reference is also classified

as a self citation in case one of the cited authors is identical to a manuscript author. Self-citation is a typical feature of scholarly work, but a high density of self-citation can be an indication of efforts to artificially increase citation levels. The use of threshold-based scoring is used when detecting anomalous self-citation patterns.

4) Reference-Local Citation Concentration Analysis: In addition to the self-citation, CSAD analyses citation concentration bias in the reference list. The frequency distribution of the authors, venues and the year of publication is examined to understand the dominance patterns. Too many citations to a few authors/venues are identified as a possible anomaly. Temporal skew analysis is also done to identify an abnormally recent or old-fashioned reference distributions, which can reflect a focus or biased literature coverage.

5) Stylometric Consistency Analysis: CSAD includes stylometric analysis in order to address the consistency of writing styles across the sections of a manuscript. The length of the sentences, lexical diversity as well as the vocabulary patterns are extracted out of the abstract and the body of the abstract. The stylometric distance measures are calculated in order to detect the dramatic changes in the writing style, which can suggest patchwork authorship, hidden content reuse, or mismatched sources of text generation. The analysis is performed on only the content of the sections but not references in order to prevent bias.

6) Explainable Risk Scoring: The result of the analysis of the sub-analyses of a CSAD is not a decision but an interpretable indicator. Weighted heuristics are used to get a composite CSAD score. CSAD also produces human-readable flags indicating the detected anomalies, such as too much self-citation, retracted citation, concentration bias in citation, or inconsistency in the stylometry. This explicable output allows editors and reviewers to know why a manuscript has a high-risk of integrity.

B. Semantic Integrity & AI-Authorship Verifier (SAIV)

Semantic Integrity & AI-Authorship Verifier (SAIV) is created to assess the ANISE across the units of the manuscript and determine the probability that a text was produced by an AI.

In comparison with superficial similarity detection methods, SAIV emphasizes more on the deeper semantic context and pattern of writing with the help of learning-based natural language processing algorithms.

$$\text{Sim}(i, j) = \frac{\mathbf{e}_i \cdot \mathbf{e}_j}{\|\mathbf{e}_i\| \|\mathbf{e}_j\|} \quad (1)$$

Transformer-based semantic similarity captures discourse-level and contextual alignment rather than strict topical equivalence. As a result, manuscripts with divergent topics but consistent academic framing may still exhibit moderate similarity values, which are conservatively flagged using predefined thresholds.

1) *Text Preprocessing and Segmentation*: After the extract of the documents, the text of the manuscript is divided into the essential sections that are the abstract, the introduction, the main body, and the conclusion where present. Basic steps of preprocessing including tokenization, normalization and elimination of non-informative artifacts are used. This provides the consistency of downstream semantic analysis and maintains original meaning of the content.

The method comprises two tasks: generating semantic embeddings and encoding semantic features to generate similar sentences.

2) *Semantic Embedding Generation*: The technique has two stages: semantic embedding generation and encoding semantic features to generate similar sentences.

To extract the semantic meaning of a scientific text, SAIV uses a transformer-based language model that is trained on academic texts. The content of every manuscript section is coded into dense vectors which are semantically rich beyond lexical similarity. These embeddings allow comparing semantic content between sections, which allows the system to determine if various sections of the manuscript are conceptually consistent.

3) *Semantic Coherence Analysis*: Semantic coherence is analyzed using similarity scores between the embedding of various sections of the manuscript. Similarity measurements can be done in pairs to determine the extent of matching between the abstract and main body and the introduction and conclusion. Low semantic similarity can be a sign of inconsistency in the flow of the narrative, change of topic, or content generated by automated text generation systems. These coherence scores are part of the total SAIV risk assessment.

4) *Probability of AI-Authorship Estimation*: Besides the coherence examination, SAIV approximates the probability of a manuscript to have AI-written or AI-aided text. A trained supervised classification model is based on scientific text embeddings to make a distinction between human and machine written materials. Instead of providing a binary label, the classifier provides a probabilistic score, and it represents uncertainty and does not imply any accusations. This likelihood is considered as a risk factor and not a factual one.

5) *Stylometric Feature Integration*: To enhance robustness, SAIV incorporates selected stylometric attributes—such as sentence length variation and lexical diversity—exclusively

within the AI-authorship classification module. These features capture subtle writing behavior patterns that may not be fully represented in embedding space. Stylometric features are not used in semantic coherence computation, thereby preventing stylistic bias in semantic similarity analysis. This separation ensures reliable AI-authorship estimation across different writing styles and academic domains.

6) *Explainable SAIV Output*: SAIV allows generating an understandable result as a set of semantic coherence scores, AI authorship probability, and descriptive flags with potential problems. Instead of indicating that the content is AI-generated, the module shows a high level of risk and indicators. These products will be used to supplement the CSAD results and give editors practical information about the integrity of the manuscripts.

C. Computation of Integrity Risk Score

The Integrity Risk Score combines signals from both CSAD and SAIV into a single, interpretable value. Rather than making a binary decision, the system uses a weighted aggregation to reflect the relative contribution of citation-based and semantic-based integrity indicators:

$$\text{Integrity Risk Score} = \alpha S_{\text{CSAD}} + \beta (1 - S_{\text{sem}}) + \gamma S_{\text{AI}}$$

$$\alpha + \beta + \gamma = 1, \quad \alpha, \beta, \gamma \in [0, 1]$$

Here, S_{CSAD} denotes the normalized citation and stylometric anomaly score, S_{semantic} represents the semantic similarity between manuscript sections, and S_{AI} corresponds to the estimated likelihood of AI-generated text. Semantic similarity is converted into a risk component using $(1 - S_{\text{semantic}})$, such that higher coherence implies lower risk.

A weighted linear aggregation is used to maintain interpretability and to model citation anomalies, semantic inconsistency, and AI-authorship likelihood as independent contributors to integrity risk. Greater emphasis is placed on citation-based indicators due to their relative robustness across research domains, while semantic and AI-based signals provide complementary contextual evidence.

Methodology Summary:

CSAD and SAIV can be used together to create a hybrid AI/Data Science approach to analyze citations through statistical analysis, check stylometric consistency, and semantic modeling with transformers. Through signal complementation and focus on explaining decisions rather than binary ones, VERITAS offers a feasible and trustworthy method of proactive research integrity evaluation.

V. IMPLEMENTATION

This part explains how the VERITAS platform is actually implemented in practice with software stack and integration of modules and its execution workflow. Its implementation is focused on modularity, reproducibility, and the ease of deployment as it is consistent with the editorial use cases in the real world.

A. Technology Stack

VERITAS is carried out in a technology stack of modern web-based technology. The Python language is used to develop the backend services and make use of its robust ecosystem of natural language processing and data analysis. The fundamental AI/Data Science modules have been applied to open-source libraries that are widely used to guarantee reliability and extensibility.

The technologies employed in the system are:

FastAPI Backend API development and orchestration.

Windows 7 PDF text and metadata extractor (PyMuPDF).

Machine learning models and statistical analysis: scikit-learn.

Transformer-based semantic embeddings Transformer Hugging Face Transformers.

SciBERT in domain-specific scientific text representation.

Pandas and NumPy to perform data processing and feature calculations.

The frontend interface is a web-based interface in the form of a dashboard, which interacts with the backend via RESTful APIs.

B. CSAD Module Implementation.

The CSAD module is used as a standalone backend API, which works on structured text and reference metadata mapped out of uploaded manuscripts. Reference parsing programs are used to obtain author names, year of publication, venue, and DOIs where possible. Author names are standardized to minimize the formatting errors in a comparison.

The retraction detection is performed using a curated retraction list, in which extracted DOIs are cross-referenced. Self-citation density: This is calculated by comparing the normalized manuscript writers and cited writers. The stylometric features of sentence length distribution and lexical diversity are also obtained based on the content sections and contrasted to determine the stylistic discrepancies. CSAD checks generate interpretable indicators all of which are rolled into a composite score.

C. SAIV Module Implementation.

Transformer-based NLP models are applied to implement the SAIV module to carry out semantic analysis. Pretrained SciBERT models are used to encode the manuscript sections into dense vectors. Cosine similarity is used to compute scores of semantic similarity between key sections.

In the case of AI-authorship estimation, SAIV makes use of a supervised classification pipeline that is trained on embedding-based feature representations. The classifier does not provide binary labels but probabilistic scores. To enhance the robustness, selected stylometric feature is combined with semantic embeddings. All the outputs are sent back in form of organized results with descriptive indicators.

D. Backend Orchestration and Data Flow.

The backend coordinates the implementation of CSAD and SAIV at the same time. When the manuscript is submitted,

the system extracts and preprocesses the text, calls the two modules of analysis and combines them to form a single Integrity Risk Score. Intermediate indicators and end reports are placed in a specific database to be able to retrieve and audit them.

The design is designed in modules so that they can be updated or extended separately. The system is currently operated in a synchronous manner so that it can generate outputs deterministically and that it is much easier to recover an error.

E. Deployment Considerations.

VERITAS can be deployed on both cloud and local environments. The modular design allows containerization of components of analysis, allowing them to be deployed and isolated scaleably. Although the present usage is limited to individual manuscript processing, the structure of the system is such that further improvements, like batch processing and cross-manuscript analysis, can be made in the future.

VI. EXPERIMENTAL SETUP

A. Dataset Preparation

Experiments were carried out to test the functionality of the suggested VERITAS platform with a mixture of real academic manuscripts and artificial scientific materials. Human-written manuscripts were represented as real research papers were gathered in open-access repositories in the realms of computer science and interdisciplinary ones.

In order to generate integrity violations in a controlled way, synthetic text samples were created with large language models and manually edited to appear as academic writing styles. These artificial samples were considered mainly to test the Semantic Integrity & AI-Authorship Verifier (SAIV) module. In the case of Citation & Stylometry Anomaly Detector (CSAD), citation-based anomalies were the products of tampering with reference lists of manuscripts by adding excessive self-citation, broken or missing Digital Object Identifiers (DOI), disproportionate years of publication, and dominance of a single author.

All papers were turned into PDFs and run through PyMuPDF to find structured text snippets, such as the abstract, body, and the reference sections, as well as the bibliographic metadata.

Figure 3 is a summarization of the frequency statistics of response length of human-written and AI-generated texts of the SAIV dataset. The variability in human written materials is higher and the mean response is also bigger, whereas the length distributions of AI-written text are more homogeneous, encouraging the application of stylometric characteristics in the estimation of AI-text authorship.

B. Experimental Protocol

The two core modules of VERITAS analyzed each manuscript independently:

The CSAD module analyzed the metadata on citations and stylometric characteristics in order to detect abnormal

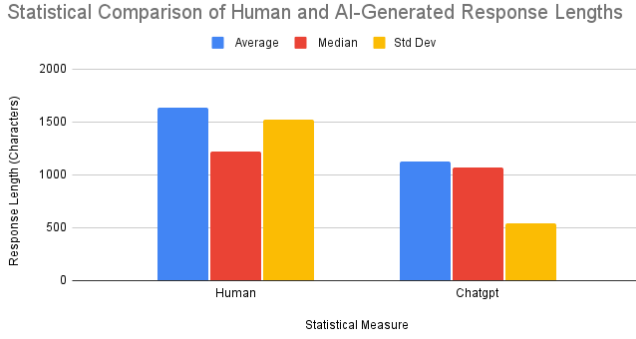


Fig. 3. Descriptive statistics of response length in the SAIV dataset for human-written and AI-generated texts, showing average, median, and variability characteristics.

reference behaviour and stylistic discrepancies across a single manuscript.

The SAIV module was used to assess the semantic consistency across various parts of a text and to approximate the probability of AI-written text based on transformer-based embeddings and trained classification.

A total score of the possible risk to the integrity of research was calculated by the output of both modules and the subsequent normalization and combination of the outcomes. All experiments were performed in off-line setting without use of web crawling or citation databases and this guarantees reproducibility and controlled evaluation.

C. Evaluation Metrics

The standard metrics used to evaluate the performance of the system included precision, recall, and F1-score to measure integrity risk detection effectiveness. In the case of the SAIV module, the semantic consistency was measured with the help of the cosine similarity of section-level semantic embeddings, whereas the classification probabilities were examined to determine the likelihood of AI authorship.

In the case of the CSAD module, statistical distributions of self-citation ratios, measure of reference concentration, and skew of citation over time have been considered to confirm the threshold of anomaly detection. Evaluation consisted of explainability and consistency of flagged outputs as opposed to binary classification.

D. Baseline Comparison

The proposed VERITAS system was qualitatively contrasted with the traditional plagiarism detection systems based on text-level similarities in the first place. In contrast to this type of systems, VERITAS was capable of revealing more profound integrity dangers, such as citation abnormalities, semantic mismatch between different parts of the manuscript and cues to AI-generated text, which are not normally considered by overlap-based detection methods.

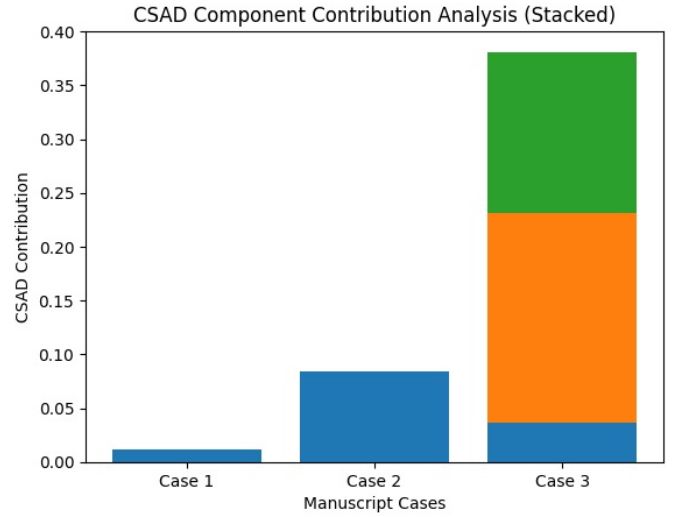


Fig. 4. Weighted contribution of self-citation ratio, retracted reference ratio, and citation density to the final CSAD score across representative manuscript cases.

VII. RESULTS AND ANALYSIS

A. CSAD Module Analysis

The Citation Stylometry Anomaly Detector (CSAD) was tested on a collection of academic texts having unaltered and synthetically written reference lists. The module was able to detect several types of irregularities in citation, such as overuse of self-citation, absence or improper structure of DOI, imbalanced reference temporal distributions, and concentration of authorship in reference lists.

It is due to the fact that manuscripts having artificially inflated ratios of self-citation may always raise red flags of high risks, whereas papers with balanced and diverse ratios of citation got lower CSAD scores. The comparison of abstract and body passages by stylometric analysis showed that documents that had been manually injected with stylistic variations have a quantifiable change in the distribution of sentence length and lexical diversity, which confirms the validity of checks made by stylometry.

Instead of generating binary decisions, CSAD was able to come up with interpretable flags on the type of anomalies detected, making it clear to see how citation behaviour is inspected.

Figure 4 illustrates the component-wise contribution to the final CSAD score. Case 1 demonstrates minimal influence from all components, resulting in a low anomaly score. Case 2 shows increased contribution primarily from self-citation, leading to a moderate risk classification. In contrast, Case 3 is predominantly influenced by the retracted reference component with an additional density penalty, producing the highest CSAD score. The visualization confirms that retracted references have the strongest impact on anomaly escalation, consistent with the assigned weighting scheme.

TABLE I
CSAD EVALUATION RESULTS ON REPRESENTATIVE MANUSCRIPT CASES

Case	Description	CSAD Score	Self-Cite Ratio	Retracted Ratio	Density (/1000w)	Stylometric Flags	Final Verdict
1	Clean Manuscript	0.14	0.06	0.00	3.2	None	Low Risk
2	Self-Citation Dominant	0.31	0.42	0.00	4.8	None	Moderate Risk
3	Retracted + Density Anomaly	0.78	0.18	0.30	9.1	Long sentences, Style shift	High Risk

TABLE II
SAIV EVALUATION RESULTS ON REPRESENTATIVE MANUSCRIPT CASES

Case	Description	Semantic Similarity	Semantic Flag	AI Likelihood	AI Flag	Notes / Explanation
1	Coherent Human-Written Manuscript	0.714	No	0.836	Yes	Moderate AI likelihood caused by structured academic style with some natural variation in phrasing.
2	Topic-Inconsistent Manuscript	0.621	Yes	0.496	No	Low AI likelihood; text shows noticeable stylistic variation typical of human authorship. Semantic inconsistency flagged.
3	Fully AI-Generated Manuscript	0.693	No	0.900	Yes	High AI likelihood due to consistently formal academic tone, dense technical phrasing, and uniform sentence structure, which are patterns commonly observed in AI-generated scholarly text.
4	Mixed-Risk Manuscript	0.650	No	0.682	No	Moderate AI likelihood caused by structured academic style with some natural variation in phrasing. No flags triggered.

B. SAIV Module Analysis

SAIV generates two scores that are complementary (i) semantic coherence scores across major manuscript sections and (ii) a probabilistic AI-authorship risk score. Notably, large AI likelihood measures are regarded as signs of stylistic risk instead of the proof of AI-generated text. The module of semantic similarity results is not biased by including stylometric features, which is only included in the AI-authorship classification module. The summary of typical cases of SAIV evaluation as presented in Table 2 shows that semantic integrity and AI-authorship risk behave independently under various conditions in a manuscript

Observations:

- **Case 1 (Human-written):** High AI likelihood shows that structured academic prose can sometimes resemble AI-generated text.
- **Case 2 (Topic-inconsistent):** Semantic flag correctly identifies mismatch between body and conclusion; AI flag not triggered.
- **Case 3 (Fully AI-generated):** AI flag accurately identifies AI-origin; semantic flag not needed since text is internally coherent.
- **Case 4 (Mixed-risk):** Moderate AI likelihood without flags demonstrates a borderline scenario where academic-

style text may appear slightly AI-like but doesn't trigger SAIV warnings.

C. Integrity Risk Score

Final Integrity Risk Score was estimated as a sum of normalized output of CSAD and SAIV. Manuscripts with both citation anomalies and semantic inconsistencies were attributed with higher overall risk scores with papers with consistent writing style and well distributed references having lower risk values.

Notably, the system was explainable as it showed individual contributing factors and the final score. This design option helps in making informed human decisions as opposed to automated rejection as it adheres to ethical standards of editorial decision-making.

D. Qualitative Comparison with Existing Tools

The conventional plagiarism detecting services are more or less superficial when it comes to text overlap and do not address the underlying issues of integrity. VERITAS in comparison showed a capability of detecting non-plagiarism-related risks, including abnormal citation patterns and semantic misalignment as would otherwise be ignored by other similarity-based systems.

These findings imply the possible potential of multi-dimensional integrity analysis as the supplementary method to the current plagiarism detection solutions.

VIII. CONCLUSION AND FUTURE WORK

The paper has introduced VERITAS, a research integrity analysis machine that proactively and correctly analyzes the scientific manuscripts and uses AI to analyze scientific research. Compared to the conventional plagiarism detection tools, which generally consider the superficial similarity of texts as the major criterion, VERITAS is a multi-dimensional tool that incorporates the citation behavior, stylometric consistency, semantic coherence, and AI-authorship probability detection.

The suggested system integrates two fundamental modules, such as the Citation and Stylometry Anomaly Detector (CSAD) and the Semantic Integrity and AI-Authorship Verifier (SAIV). Experimental testing established that CSAD has been shown to be effective in detecting irregularities in citation, including excessive citing of oneself, concentration bias in references and stylistic discrepancies, whereas SAIV detects semantic mismatch and AI indicative probabilities. Notably, the system places an emphasis on interpretability by offering explainable flags and contributing factors and a general score of the Integrity Risk, which will help to make well-informed human decisions instead of automated rejection of information.

Although the implementation currently under consideration is based on paper-internal and reference-local analysis performed in the offline setting, there are multiple ways to improve the approach in the future. The network-level citation analysis of a large number of manuscripts can be included to capture large scale citation rings and coordinated citation behavior. Further robustness may be obtained by extending to multilingual manuscripts and adversarially paraphrased manuscript. In addition, by incorporating VERITAS into editorial routines and submission systems, one could permanently validate integrity assessment parameters and refine them.

VERITAS can also serve as a viable and scalable solution to improve trust and accountability in scholarly publishing because it focuses on the attributes of transparency, modularity, and ethical use of AI.

REFERENCES

- [1] S. Kojaku, G. Livan, and N. Masuda, "Detecting anomalous citation groups in journal networks," *Scientific Reports*, vol. 11, Art. no. 14524, 2021, doi: 10.1038/s41598-021-93572-3.
- [2] J. Liu, F. Xia, X. Feng, J. Ren, and H. Liu, "Deep graph learning for anomalous citation detection," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 33, no. 6, pp. 2543–2557, Jun. 2022, doi: 10.1109/TNNLS.2022.3145092.
- [3] J. D. Wren and C. Georgescu, "Detecting anomalous referencing patterns in PubMed papers suggestive of author-centric reference list manipulation," *Scientometrics*, vol. 127, pp. 5753–5771, 2022, doi: 10.1007/s11192-022-04503-6.
- [4] C. Candal-Pedreira, C. Guerra-Tort, A. Ruano-Ravina, F. Freijedo-Fariñas, J. Rey-Brandariz, J. S. Ross, and M. Pérez-Ríos, "Retracted papers originating from paper mills: A cross-sectional analysis of references and citations," *Journal of Clinical Epidemiology*, vol. 172, Art. no. 111397, 2024, doi: 10.1016/j.jclinepi.2024.111397.
- [5] J. Liu, X. Wang, and X. Liang, "Bibliometric feature identification and analysis of retracted papers in biomedicine: An interpretable machine learning perspective," *Information Processing & Management*, vol. 62, no. 2, Art. no. 103708, 2025, doi: 10.1016/j.ipm.2024.103708.
- [6] H. Desaire, A. E. Chua, M.-G. Kim, and D. Hua, "Accurately detecting AI text when ChatGPT is told to write like a chemist," *Cell Reports Physical Science*, vol. 4, Art. no. 101672, 2023, doi: 10.1016/j.xcrp.2023.101672.
- [7] D. Weber-Wulff, A. Anohina-Naumeca, S. Bjelobaba, T. Foltýnek, J. Guerrero-Dib, O. Popoola, P. Šigut, and L. Waddington, "Testing of detection tools for AI-generated text," *International Journal for Educational Integrity*, vol. 19, Art. no. 26, 2023, doi: 10.1007/s40979-023-00146-z.
- [8] I. Beltagy, K. Lo, and A. Cohan, "SciBERT: A pretrained language model for scientific text," in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, Hong Kong, 2019, pp. 3615–3621.
- [9] M. Maral, "A bibliometric analysis on academic integrity," *Journal of Academic Ethics*, vol. 22, no. 4, pp. 665–687, 2024, doi: 10.1007/s10805-024-09519-6.
- [10] J. H. Fowler and D. W. Aksnes, "Does self-citation pay?" *Scientometrics*, vol. 72, no. 3, pp. 427–437, 2007, doi: 10.1007/s11192-007-1777-2.
- [11] J. P. A. Ioannidis, J. Baas, R. Klavans, and K. W. Boyack, "A standardized citation metrics author database annotated for scientific field," *PLOS Biology*, vol. 17, no. 8, p. e3000384, 2019, doi: 10.1371/journal.pbio.3000384.
- [12] I. Fister Jr., I. Fister, and M. Perc, "Toward the discovery of citation cartels in citation networks," *Frontiers in Physics*, vol. 4, Art. no. 49, 2016, doi: 10.3389/fphy.2016.00049.
- [13] K. Przystalski, J. K. Argasiński, I. Grabska-Gradzińska, and J. K. Ochab, "Stylometry recognizes human and LLM-generated texts in short samples," *Expert Systems with Applications*, vol. 296, Part B, Art. no. 129001, Jan. 2026, doi:10.1016/j.eswa.2025.129001. :contentReference[oaicite:0]index=0